Time in Production and the Macroeconomy

Research Proposal

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Motivation

- Supply-Chain Disruptions (Covid, Trump, etc.)
  - Focused on Networks (Production requires good from a variety of firms)
  - [Gabaix, 2011, Acemoglu et al., 2012, Carvalho, 2014, Acemoglu et al., 2017, Baqaee and Farhi, 2019, Baqaee and Farhi, 2020]
- Production requires time
  - Abstract away from Complicated Networks
  - Focus on understanding time to build.
  - [Kydland and Prescott, 1982, Antràs, 2023a, Antràs, 2023b, Antràs and Tubdenov, 2025, Leng et al., 2025]

Introduction

- Questions:
  - How does Time in Production shape the transmission of shocks?
- Answers:
  - One-time Productivity shocks can have persistence.

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Setting

- Discrete-time
- No capital
- Representative agent households
- Final Good firm with CES structure
- perfectly competitive intermediate goods firms with heterogeneous time in production productivity

Households

- Assume that the household sector faces the following problem:

$$\max \sum_t \beta^t \left(\theta_t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \chi \frac{L_t^{1+\eta}}{1+\eta} \right) \right)$$

subject to

$$P_t C_t = W_t L_t$$

Final Goods

- Final Goods Production Function

$$Y_\tau = \left(\sum_{i=0}^{F} (\tilde{y}_{i,\tau})^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}}$$

- $\tilde{y}_{i,\tau}$ is the demand at time τ for intermediate goods of type i
- $\tilde{y}_{i,\tau} = \sum_{t \leq \tau} y_{i,\tau|t}$, where $y_{i,\tau|t}$ is the amount of good i delivered at time τ starting at time t.

Intermediate Goods Producer: Production Function

- Intermediate Goods Production Function:

$$y_{i,\tau|t} := \kappa^{\tau-t} \left(\tau - t + B_t N_i\right) \left(A_t l_{t,i}\right)$$

- $\kappa \in (0,1)$- physical cost of time to build.
- $N_i \in \mathbb{R}$ - Firm specific time-to-build productivity

Intermediate Goods Producer: Problem

- A producer of intermediate good i solves the following lifetime discounted problem which is

$$\max_{t \leq \tau_t \leq T_i + t, l_t} \sum_{t \geq 0} \left(p_{i,\tau_t} \prod_{j=0}^{\tau_t} \delta_j y_{i,\tau_t|t} - \prod_{j=0}^{t} \delta_j W_t l_t \right)$$

- $p_{i,t}$ - price of good i at time t
- W_t - wage at time t

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Equilibrium I

Definition

(Endogenous Equilibrium): Given initial conditions$\{\{y_{i,0|t}\}_{i=0}^{F}\}_{t=-\infty}^{0}$, and given a sequence of interest rates $\{i_t\}_{t=0}^{\infty}$ an equilibrium is a sequence of prices $\{P_t, W_t, \{p_{i,t}\}_{i=0}^{F}, \}_{t=0}^{\infty}$, and allocations $\{C_t, Y_t, L_t, \{I_{i,t}, \tilde{y}_{i,t}, \{y_{i,t|s}\}_{s=0}^{t}, \tau_{i,t}\}_{i=0}^{F}\}_{t=0}^{\infty}$

- Households maximize utility given prices prices $\{P_t, W_t, i_t\}$, and choose $\{C_t, L_t\}$
- Final good produces maximizes profits, given prices $\{P_t, p_{i,t}\}$, and choose $\{Y_t, \{, \tilde{y}_{i,t|s}\}\}$ for all t
- A firm of type i at time t maximizes profits given prices $\{i_t, W_t, \{p_{i,s}\}_{s\geq t}\}$,and choose $\{I_{i,t}, y_{i,\tau_{i,t}|s}, \tau_{i,t}\}$.

Equilibrium II

Definition

The following market-clearing conditions

- $C_t = Y_t$ for all t
- $L_t = \sum_{i=0}^{F} l_{i,t}$ for all t
- $\tilde{y}_{i,t} = \sum_{s \leq t} y_{i,t|s}$ for all t and s

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FOCS: SS

- At the SS, choice of τ will be the same, we have that
- τ FOC:

$$\tau_{i,t} - t = \max\left(-\frac{1}{\log(\delta\kappa)} - BN_i, 0\right)$$

  - $\delta\kappa$ increases $\tau_{i,t}$
  - N_i decreases $\tau_{i,t}$
- Effective Productivity: $z_i(\tau|t) = (\kappa\delta)^{\tau-t}(\tau - t + B_t N_i) A_t$

Small Shocks

Proposition

Given a sequence $\{i_s, A_s, B_s\}$ *such that for all* s, i_s, A_s, B_s *are constant, expect at time* t. *Then there* $A_t, B_t, i_t \neq A_s, B_s, i_s$ *such that* $\tau_{is} = \tau_{is}^{ss}$.

Small Shocks

- Conjecture that we have an equilibrium where τ_{it} is the same as in the SS
- Compute prices. and show that price changes are small.
- Results for Endogenous Time to Build and Exogenous Time to Build are the same.

First Order Approximation

Definition

Given $\{\{\tau_{i,t}\}_{i=0}^{F}\}_{t=0}^{\infty}$ then $M_{t+k|t} := \{i|\tau_{i,t} = t+k\}$

Corollary (Shocks on GDP)

Suppose there is an infinitesimally small, one-time shock at time τ around the steady state then the impact on GDP $Y_{\tau+j}$ for $j \geq 0$ is

- *Productivity Shock:* $\frac{\partial \log Y_{\tau+j}}{\partial \log A_t} = \frac{\eta+1}{\eta+\sigma} \frac{\sum_{i \in M_{\tau+j|\tau}} (z_i(\tau+j|\tau))^{\varepsilon-1}}{\sum_i (z_i(\tau+j|s_i))^{\varepsilon-1}}$
- *Interest Rate Shock for $j > 0$:*
  $\frac{\partial \log Y_{\tau+j}}{\partial \log \beta_t} = \frac{\eta+1}{\eta+\sigma} \frac{\sum_{i \in M_{\tau+j|\tau}} (z_i(\tau+j|t))^{\varepsilon-1}}{\sum_i (z_i(\tau+j|s_i))^{\varepsilon-1}}$
- *Interest Rate Shock for $j = 0$:* $\frac{\partial \log Y_\tau}{\partial \log \beta_t} = 0$

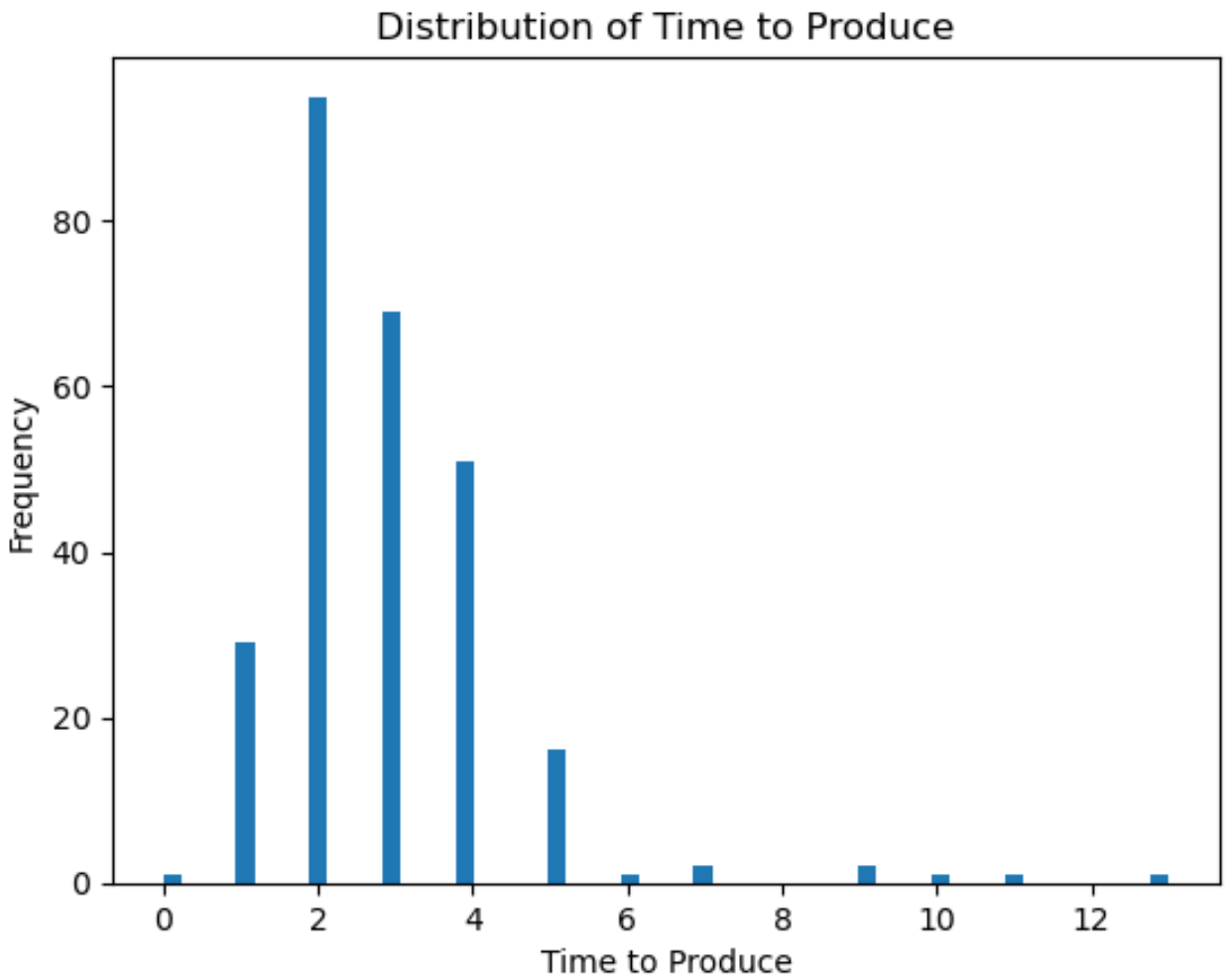
Stylized Example

Lemma (Stylized Illustration of Shocks)

Suppose that $z_i(\tau|t) = \bar{z}$ for all i, and let λ be the counting measure, then under the SS following are true

- *Productivity Shock:* $\dfrac{\partial \log Y_{t+j}}{\partial \log A_t} = \dfrac{\lambda\left(M_{t+j|t}\right)}{F+1}$
- *Rate Shock $j \neq 0$:* $\dfrac{\partial \log Y_{t+j}}{\partial \log \beta_t} = \dfrac{\lambda\left(M_{t+j|t}\right)}{F+1}$

τ_i-Distribution ([Antràs and Tubdenov, 2025])



Productivity (A) Shock Propagation

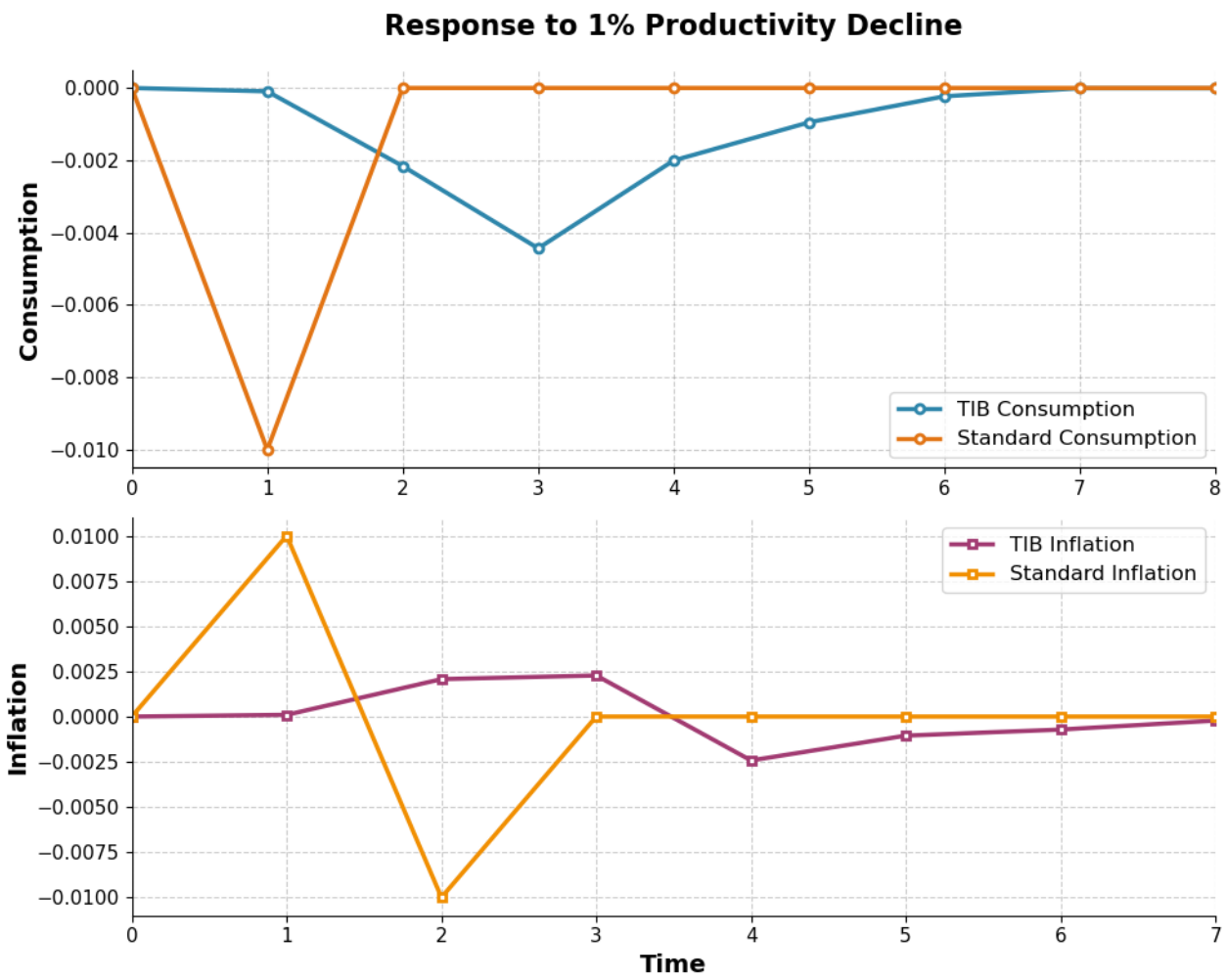
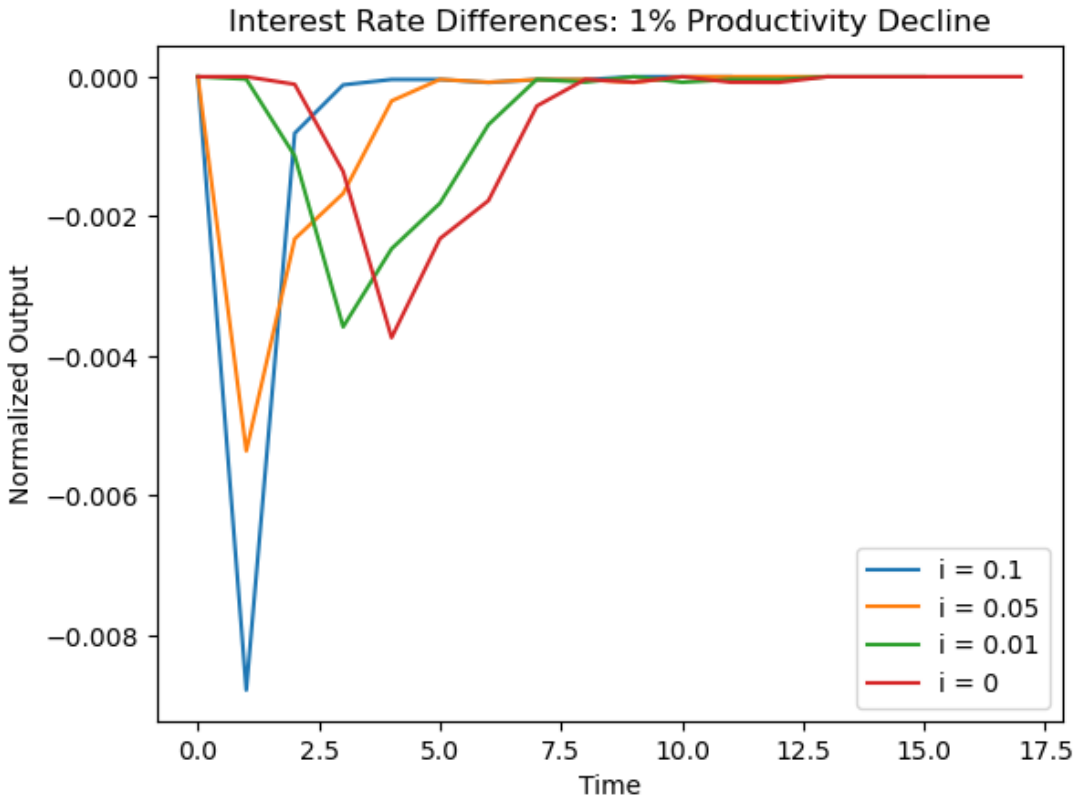
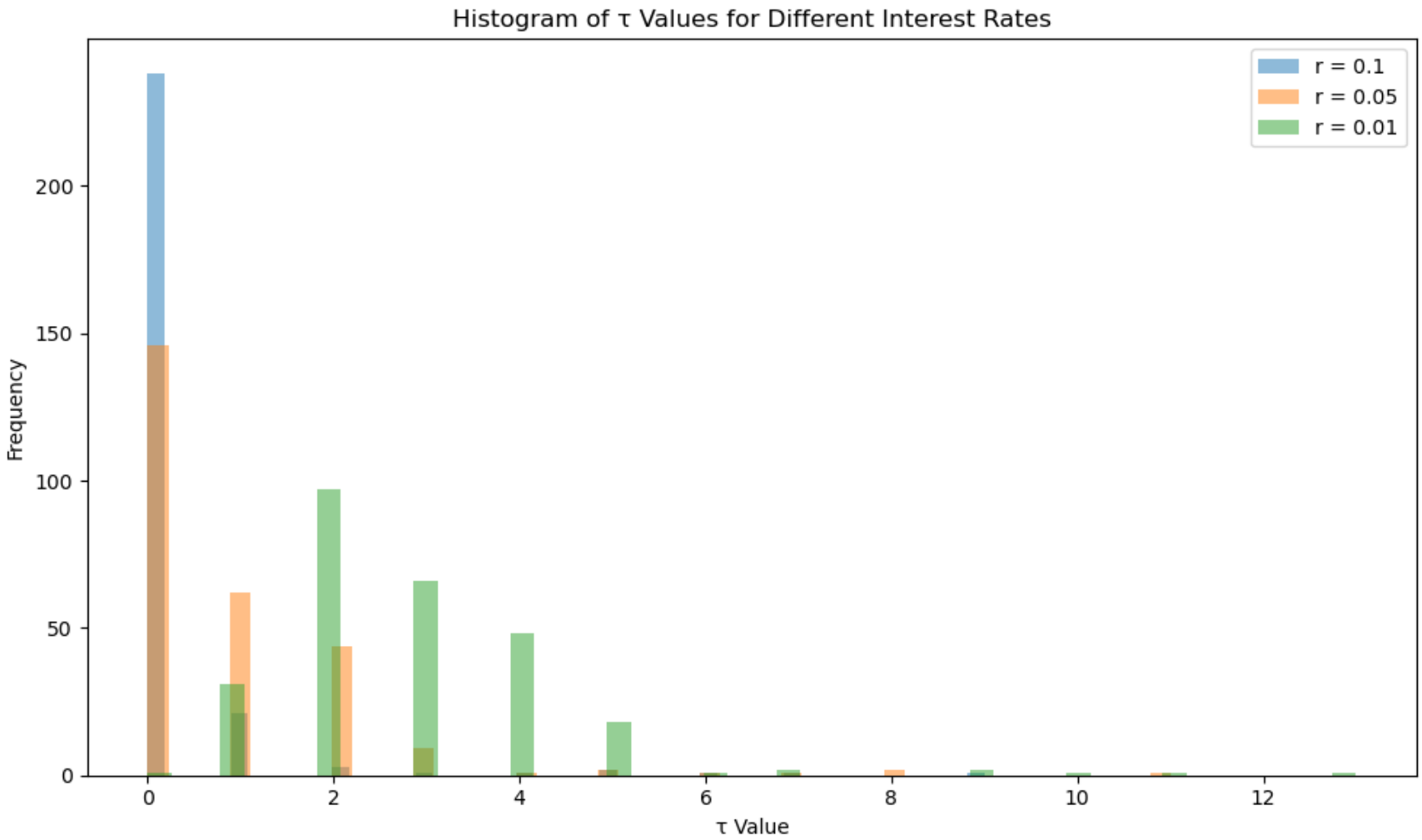


Figure 1: Real Model

Monetary Policy Environment



Histogram of τ under different interest rates



Complementarity

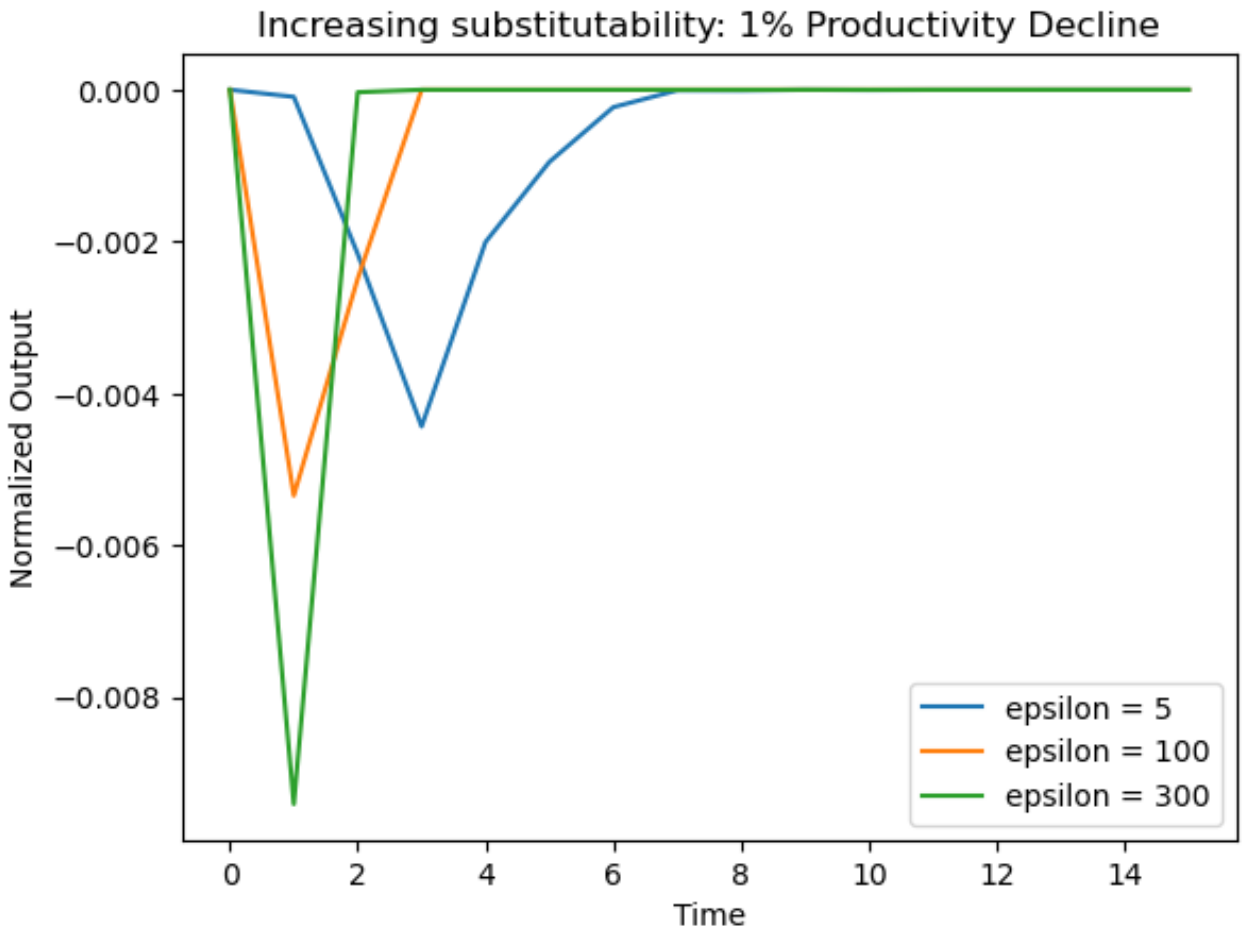


Figure 2: Effect of substitutability parameter ε

Intuition: Complementarity

Lemma

Suppose that there exists a unique maximum z^* associated with firms of type i. Then, around the SS, it follows that

- *Suppose that* $i \notin M_{t+k|t}$ *then* $\lim_{\varepsilon \to \infty} \frac{\partial \log Y_{t+k}}{\partial \log A_t} = 0$
- *Suppose that* $i \in M_{t+k|t}$ *then* $\lim_{\varepsilon \to \infty} \frac{\partial \log Y_{t+k}}{\partial \log A_t} = 1$

Growth

Lemma

Let $\tau_{i,t}^{SS}$ be under the SS, and let $\tau_{i,t}^{g}$ be the time to build under growth rate g ($A_t = (1+g)^t A_0$) then

- *if $g > 0$, $\tau_{i,t}^{SS} \geq \tau_{i,t}^{g}$*
- *if $g < 0$, $\tau_{i,t}^{SS} \leq \tau_{i,t}^{g}$*

Intuition (Positive Growth)

- Under RE, firms perceive prices going down
- Incentives firms to reduce time to build.
-

$$\tau_{i,t}^{g} - t = \max\left(-\frac{1}{\log\left(\delta\kappa/(1+g)\right)} - BN_i, 0\right)$$

Conclusion

- Still a lot of work!
  - What would be the effect of policy?
- Thank you for showing up!

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